Practice Topics > CFA-26-11-LII-B: Financial … > Module 2: Employee Compens…

Review

60%

18
Out of 30 points

1:57
Avg Time Per Question

Your Answers:

Vignette

Kensington plc, a fictional company based in the United Kingdom, sponsors a DB pension plan for qualifying employees. Kensington prepares its financial statements under IFRS. The discount rate that the company used in estimating the present value of its pension obligation was 5.48%. Disclosures on Kensington's pension plan in the company's notes to financial statements for the year ended 31 December 20X1 included the following.

Components of periodic benefit cost	
Service cost	£228
Net interest (income) expense	273
Remeasurements	−18
Periodic pension cost	£483

Change in benefit obligation	
Benefit obligations at beginning of year	£28,416
Service cost	228
Interest cost	1,557
Benefits paid	−1,322
Actuarial gain or loss	0
Benefit obligations at end of year	£28,879

Change in plan assets

Fair value of plan assets at beginning of year	£23,432
Actual return on plan assets	1,302
Employer contributions	693
Benefits paid	−1,322
Fair value of plan assets at end of year	£24,105
Funded status at beginning of year	**−£4,984**
Funded status at end of year	**−£4,774**

1 Multiple Choice 1 / 1 point

At 31 December 20X1, GBP 28,879 million represents:

A. the funded status of the plan.

B. the DB obligation.

C. the fair value of the plan's assets.

Feedback

B is correct. GBP 28,879 million is the present value of future benefits as at 31 December 20X1. This is the "gross" liability, before netting the fair value of plan assets to calculate the funded status.

2 Multiple Choice 1 / 1 point

The GBP 1,284 million difference in interest expense reported on the income statement and the interest cost on the benefit obligation in 20X1 is a result of:

A. interest income on plan assets.

B. the actual return on plan assets.

C. different assumed discount rates.

Feedback

A is correct. The interest expense reported on the income statement is a “net” interest expense/income amount computed as the product of the funded status and discount rate at the beginning of the year. Equivalently, it is the (discount rate x benefit obligation) – (discount rate x fair value of plan assets).

---

3 Multiple Choice 0 / 1 point

The amount recognized by Kensington as an operating expense on the income statement for the year ended 31 December 20X1 is closest to:

A. 210.

Correct Answer: **B. 228.**

B. 228.

C. 483.

Feedback

B is correct. Service cost is an operating expense, representing the increase in the benefit obligation from current and past service. Net interest expense/income is financing expense/income recognized below the operating income line. Remeasurements are recognized in OCI, not in earnings.

---

4 Multiple Choice 1 / 1 point

The cash outflow recognized by Kensington in cash flows from operating activities for the year ended 31 December 20X1 is closest to:

A. 228.

B. 693.

C. 1,322.

Feedback

B is correct. Employers' plan contributions are cash outflows in operating activities.

5 Multiple Choice 1 / 1 point

The amount recognized on the balance sheet decreased from 31 December 20X0 to 31 December 20X1 because:

A. The sum of service cost and interest cost exceeded benefits paid.

B. the discount rate used in estimating the pension obligation exceeded the actual rate of return of plan assets for the year.

C. the sum of the actual return on plan assets and employer contributions exceeded the sum of service and interest cost on the benefit obligation.

Feedback

C is correct. The net pension liability recognized on Kensington's balance sheet decreased because the fair value of plan assets increased by more than the benefit obligation.

A is incorrect. While it is true that the sum of service and interest costs exceeded benefits paid, benefits paid is deducted from both the benefit obligation and plan assets, so the relevant spread is between the sum of service and interest costs and the sum of return on plan assets and employer contributions.

B is incorrect. The discount rate used in estimating the pension obligation was 5.48% and the actual rate of return on plan assets was 1,302/23,432 = 5.56%, 8 basis points higher, not lower.

6 Multiple Choice 1 / 1 point

An analyst preparing a discounted cash flow model on 14 January 20X2 to value Kensington's equity should deduct which of the following from the estimate of enterprise value to arrive at equity value?

A.
4,774

B. 4,984

C. 28,879

Feedback

A is correct. The analyst should deduct the net pension liability as of 31 December 20X1 from the estimate of enterprise value to arrive at equity value, as if it were debt.

B is incorrect. This is the net pension liability from the beginning of 20X1.

C is incorrect. 28,879 is the gross benefit obligation. Deducting this from enterprise value to arrive at equity value would ignore plan assets that are exclusively for the payment of benefits to plan beneficiaries.

7 Multiple Choice 1 / 1 point

A 100 basis point decrease in investment grade corporate bond yields may affect Kensington's plan funded status by less than the increase in the benefit obligation because:

A. remeasurements from changes in assumptions are recognized in OCI, not in earnings.

B. a decrease in service cost will partially offset the increase.

C. the fair value of plan assets may simultaneously increase.

Feedback

C is correct. A decrease in interest rates may result in an increase in the fair value of plan assets that offsets the increase in the benefit obligation from a lower discount rate, especially plan assets invested in longer-duration fixed-income securities.

Vignette

XYZ SA is a fictional company that uses a DB pension plan and stock option grants as part of its compensation to qualified employees. XYZ SA prepares its financial statements under IFRS.

Information on XYZ's DB plan and volatility assumptions used to value stock option grants were as follows:

XYZ SA Defined BP Information, Fiscal Year 2024	
Employer contributions	1,000
Current service costs	200

XYZ SA Defined BP Information, Fiscal Year 2024

Past service costs	120
Discount rate used to estimate plan liabilities at beginning of year	7.00%
Benefit obligation at beginning of year	42,000
Benefit obligation at end of year	41,720
Actuarial loss due to increase in plan obligation	460
Plan assets at beginning of year	39,000
Plan assets at end of year	38,700
Actual return on plan assets	2,700
Expected rate of return on plan assets	8.00%

Grant Year	**Weighted Average Expected Volatility**
2024 Valuation Assumptions	
2020–2024	21.50%
2023 Valuation Assumptions	
2019–2023	23.00%

8 Multiple Choice 1 / 1 point

The amount recognized by XYZ as operating expense on the income statement related to its DB plan for fiscal year 2024 is closest to:

A. 200.

B. 320.

C. 1,000.

Feedback

B is correct. Service cost, comprising current and past service costs of 200 and 120, respectively, is recognized on the income statement as an operating expense.

9 Multiple Choice 0 / 1 point

If XYZ prepared its financial statements under US GAAP, the total amount recognized by XYZ on the income statement related to its DB plan for fiscal year 2024 (assuming the company chooses not to immediately recognize the actuarial loss and assuming there is no amortization of past service costs or actuarial gains and losses) would be closest to:

A. 20.

B. 59.

Correct Answer: **A. 20.**

C. 530.

Feedback

A is correct. Under US GAAP—assuming the company chooses not to immediately recognize the actuarial loss and assuming there is no amortization of past service costs or actuarial gains and losses—the components of periodic pension cost that would be reported in P&L include the current service cost of 200, the interest expense on the pension obligation at the beginning of the period of 2,940 [= 7.0% × 42,000], and the expected return on plan assets, which is a reduction of the cost of 3,120 (= 8.0% × 39,000). Summing these three components gives 20.

10 Multiple Choice 1 / 1 point

An analyst is building a financial statement model for XYZ SA. The analyst assumes that service cost and the discount rate in FY2025 will be the same as in the previous year. The analyst's estimate of pension cost recognized on the income statement in FY2025 is closest to:

A. 320.

B. 404.

C. 531.

Feedback

C is correct. Pension costs recognized on the income statement comprise service cost and net interest expense/income. The analyst assumes service cost remains the same as in the previous year, or (200+120) = 320. Net interest expense/income is the product of the discount rate and the net pension liability/asset at the beginning of FY2025, or the end of FY2024, [(41,720-38,700) x 0.07] = 211. Summing these two components gives 531.

11 Multiple Choice 1 / 1 point

If XYZ had used the same volatility assumption for its FY2024 option grants that it had used in FY2023, its FY2024 net income would have been:

- A. lower.
- B. higher.
- C. the same.

Feedback

A is correct. In FY2024, XYZ used a lower volatility assumption than it did in FY2023. Lower volatility reduces the fair value of an option and thus the reported expense as the award vests. Using the FY2023 volatility estimate would have resulted in higher expense and thus lower net income.

12 Multiple Choice 1 / 1 point

If XYZ SA also granted RSUs to employees in fiscal 2024, the decrease in XYZ SA's share price volatility assumption would:

- A. increase the grant-date fair value of the RSUs.
- B. decrease the grant-date fair value of the RSUs.
- C. not affect the grant-date fair value of the RSUs.

Feedback

C is correct. The grant-date fair value of an RSU is the share price, which may be adjusted for expected dividends. The volatility assumption is not relevant to the valuation of RSUs.

Vignette

Sallie Kwan Industrials (SKI) is a Singapore-based automation equipment manufacturer that reports under US GAAP. The company disclosed the following information in a note to its financial statements titled "Share-Based Compensation."

Under our Share Incentive Plan, the Company grants restricted stock units ("RSUs") to its officers, employees, directors and other eligible persons of up to 83,000,000 Class A ordinary shares. RSUs vest 25% on the first anniversary year from the grant date and the remaining 75% vest in 12 substantially equal quarterly installments. RSU activity for the two years ended 31 December 20X2 was as follows.

	Number of Shares	**Weighted Average Grant-Date Fair Value per Share (SGD)**
Unvested at 31 December 20X0	4,754,972	12.34
Granted	6,249,313	20.50
Vested and settled	(2,131,415)	13.67
Forfeited	(791,433)	15.22
Unvested at 31 December 20X1	8,081,437	18.02
Granted	5,034,735	72.37
Vested and settled	(3,332,063)	19.25
Forfeited	(442,181)	28.74
Unvested at 31 December 20X2	9,341,928	46.36

Share-based compensation expense for RSUs is measured based on the fair value of the Company's ordinary shares on the date of grant. SKI accounts for forfeitures as they occur.

Unrecognized share-based compensation expense as of 31 December 20X0, 20X1, and 20X2 was SGD 58.7, SGD 145.6, and SGD 433.1 million, respectively.

13 Multiple Choice 0 / 1 point

The amount recognized as operating expense on SKI's income statement related to its Equity Incentive Plan for the year ended 31 December 20X2 is closest to:

A. SGD 51.4 million.

B. SGD 64.1 million.

× C. SGD 123.1 million.

Correct Answer: **A. SGD 51.4 million.**

Feedback

A is correct. The amount recognized as operating expense is the share-based compensation expense, which the product of 3,332,063 RSUs vested with a per-share grant-date fair value of SGD 19.25 less forfeitures of 442,181 with a per-share grant-date fair value of SGD 28.74. (3,332,063 x 19.25) – (442,181 x 28.74) = 51,433,931.

---

14 Multiple Choice 0 / 1 point

The increase in SKI's basic shares outstanding in the year ended 31 December 20X2 from its Equity Incentive Plan is closest to:

× A. 2,889,882.

Correct Answer: **B. 3,332,063.**

B. 3,332,063.

C. 5,034,735.

Feedback

B is correct. Settlement of RSUs increases basic shares outstanding. The number of RSUs settled (converted to common shares) in 20X2 is 3,332,063. Forfeit RSUs are not deducted, as these are RSUs, not common shares.

---

15 Multiple Choice 0 / 1 point

The average market price for SKI shares at the time of settlement for the RSUs that vested in 20X2 was SGD 71.50. Assuming a statutory tax rate of 17%, the impact to SKI's income tax expense for 20X2 related to share-based compensation is closest to:

A. a SGD 29.6 million increase in income tax expense.

B. a SGD 29.6 million reduction in income tax expense.

C. a SGD174.1 million reduction in income tax expense.

Correct Answer: **B. a SGD 29.6 million reduction in income tax expense.**

Feedback

B is correct. The excess tax benefit or windfall in 20X2 is equal to the statutory tax rate multiplied by the amount that the tax deduction associated with the settlement of the share-based award exceeds the share-based compensation expense recognized on the income statement: statutory tax rate x (tax deduction – share-based compensation expense). This is equal to 0.17 x [(71.50 x 3,332,063) – (19.25 x 3,332,063)] = 29,597,050.

---

16 Multiple Choice 0 / 1 point

If SKI reported under IFRS instead of US GAAP, its effective tax rate would likely be:

A. lower.

Correct Answer: **B. higher.**

B. higher.

C. the same.

Feedback

B is correct. Excess tax benefits or tax windfalls related to share-based compensation are recognized as gains directly in equity under IFRS, not as reductions in income tax expense on the P&L under US GAAP.

---

17 Multiple Choice 1 / 1 point

SKI reported basic shares outstanding of 270,400,000 and positive net income on its income statement for the year ended 31 December 20X2. Besides unvested RSUs, SKI had no other potentially dilutive securities outstanding. Assuming an average share price of SGD 71.50 for the year, diluted shares outstanding for the year ended 31 December 20X2 is closest to:

A. 273,284,585.

B. 275,695,075.

C. 279,341,928.

Feedback

B is correct. Diluted shares outstanding is computed using the treasury stock method, where unvested RSUs are added to basic shares outstanding, net of assumed repurchases which is based on the average unrecognized share-based compensation expense and average prevailing market price for the shares.

Basic shares outstanding:	270,400,000
Plus: Unvested RSUs of	9,341,928
Minus: Average unrecognized share-based compensation expense of (145.6 + 433.1 million)/2	4,046,853
= 289.350 million / average share price of 71.50 =	
Diluted shares outstanding:	275,695,075

18 Multiple Choice 0 / 1 point

If the weighted average share price during 20X2 was lower than SGD 71.50, but the average unrecognized share-based compensation expense was the same, diluted shares outstanding would be _______________ the amount calculated in the previous question.

A. Lower than

× B. Higher than

Correct Answer: **A. Lower than**

C. The same as

Feedback

A is correct. A lower weighted average share price in 20X2 would result in a greater number of shares assumed to be repurchased, thus fewer shares added to basic shares outstanding in the calculation of diluted shares outstanding.

19 Multiple Choice 0 / 1 point

An analyst is building a financial statement model for SKI and projects share-based compensation expense to be 10% of revenues next fiscal year. Based on the following assumptions, the analyst's estimated basic share count for SKI will increase by how many shares next fiscal year?

Revenues FY2025E = SGD 506 million

Weighted-average grant-date fair value per share of vesting RSUs = SGD 46.36

Forfeitures = 0 shares (the analyst's share-based compensation expense is net of forfeitures).

A. 1.1 million

B. 1.3 million

Correct Answer: **A. 1.1 million**

C. 10.9 million

Feedback

A is correct. Share-based compensation expense is estimated to be SGD 506 million x 0.10 = SGD 50.6 million. The expense is recognized upon vesting (which for RSUs is coincident with settlement). The number of shares issued and therefore increasing basic shares outstanding is SGD 50.6 million / SGD 46.36 = 1.1 million shares.

Vignette

Terra Mercado is a retailer that compensates employees with a DB pension plan and stock options in addition to short-term benefits and a DC plan. Terra Mercado prepares its financial statements in accordance with US GAAP and reports in BRL.

Peter Friedland, CFA, is an equity analyst concerned with earnings quality. He is particularly interested in whether the discretionary assumptions the company is making regarding compensation plans are contributing to the recent earnings growth at Terra Mercado. He gathers information from the company's regulatory filings regarding the pension plan assumptions and assumptions related to option valuation for the three years ended 31 December 20X3.

Assumptions Used for Terra Mercado DB Plan

	20X3	20X2	20X1
Expected long-term rate of return on plan assets	6.06%	6.14%	6.79%

Assumptions Used for Terra Mercado DB Plan

	20X3	20X2	20X1
Discount rate used to estimate pension obligation at beginning of year	4.85%	4.94%	5.38%
Estimated salary growth rate	4.00%	4.44%	4.25%
Estimated inflation rate	3.00%	2.72%	2.45%

Assumptions Used for Terra Mercado Option Grants

	20X3	20X2	20X1
Risk-free rate	4.6%	3.8%	2.4%
Expected life	5.0 yrs	4.5 yrs	5.0 yrs
Dividend yield	1.0%	0.0%	0.0%
Expected volatility	29%	31%	35%

20 Multiple Choice 1 / 1 point

Compared to the 20X3 reported financial statements, if Terra Mercado had used the same expected long-term rate of return on plan assets assumption in 20X3 as it used in 20X1, its year-end 20X3 pension obligation would most likely have been:

A. lower.

B. higher.

C. the same.

Feedback

C is correct. The assumed long-term rate of return on plan assets is not a component used in calculating the pension obligation, so there would be no change. However, the long-term rate of return can affect the pension expense under US GAAP.

21 Multiple Choice 1 / 1 point

Compared to the reported 20X3 financial statements, if Terra Mercado had used the same discount rate it used in 20X1, it would have most likely reported lower:

A. net income.

B. total liabilities.

C. cash flow from operating activities.

Feedback

B is correct. A higher discount rate (5.38% instead of 4.85%) will reduce the present value of the pension obligation (liability). In most cases, a higher discount rate will decrease the interest cost component of the net periodic cost because the decrease in the obligation will more than offset the increase in the discount rate (except if the pension obligation is of short duration). Therefore, periodic pension cost would have been lower and reported net income higher. Cash flow from operating activities should not be affected by the change.

---

22 Multiple Choice 1 / 1 point

Compared to the assumptions Terra Mercado used to compute its periodic pension cost in 20X1, earnings in 20X3 were most favorably affected by the change in the:

A. discount rate.

B. estimated future salary increases.

C. expected long-term rate of return on plan assets.

Feedback

B is correct. In 20X3, the three relevant assumptions were lower than in 20X1. Lower expected salary increases reduce the service cost component of the periodic pension cost. A lower discount rate will increase the DB obligation and increase the interest cost component of the periodic pension cost (the increase in the obligation will, in most cases, more than offset the decrease in the discount rate). Reducing the expected return on plan assets typically increases the periodic pension cost.

---

23 Multiple Choice 0 / 1 point

Compared to the pension assumptions Terra Mercado used in 20X2, which of the following pairs of assumptions used in 20X3 is *most likely* internally inconsistent?

A. Estimated future salary increases, inflation

B. Discount rate, estimated future salary increases

Correct Answer: **A. Estimated future salary increases, inflation**

C. Expected long-term rate of return on plan assets, discount rate

Feedback

A is correct. The company's inflation estimate rose from 20X2 to 20X3. However, it lowered its estimate of future salary increases. Normally, salary increases are positively related to inflation.

---

24 Multiple Choice 0 / 1 point

Compared to the reported 20X3 financial statements, if Terra Mercado had used the 20X1 volatility assumption to value its employee stock option grants, it would have most likely reported higher:

A. net income.

B. operating expense.

C. accrued compensation liability.

Correct Answer: **B. operating expense.**

Feedback

B is correct. A higher volatility assumption increases the value of the stock option and thus the compensation expense, which, in turn, reduces net income. There is no associated liability for equity-settled stock options.

---

25 Multiple Choice 1 / 1 point

Compared to the assumptions Terra Mercado used to value stock options in 20X2, earnings in 20X3 were most favorably affected by the change in the:

A. expected life.

B. risk-free rate.

C. dividend yield.

Feedback

C is correct. A higher dividend yield reduces the value of the option and thus option expense. The lower expense results in higher earnings. Higher risk-free rates and expected lives result in higher call option values.

Vignette

Your colleague is building financial statement and discounted cash flow models for a company and has asked you review their modeling decisions with respect to the subject company's post-employment and stock-based compensation plans.

26 Multiple Choice 1 / 1 point

Your colleague's discounted cash flow model expenses service cost and net interest expense in free cash flow and deducts the company's net pension liability from enterprise value to arrive at an estimate of equity value. You *most likely* recommend that your colleague:

A. make no changes.

B. remove service cost from free cash flow because it is a non-cash expense.

C. remove net interest expense from free cash flow because the net pension, at present value, is already deducted from enterprise value.

Feedback

C is correct. Net interest expense for a DB plan represents the unwinding of the discount with the passage of time. The discounted cash flow model values a company at the present time. Service cost is correctly expensed in the model as it represents increases in the pension obligation unrelated to the time value of money.

27 Multiple Choice 1 / 1 point

Your colleague's balance sheet is imbalanced, with assets exceeding total liabilities and equity by the same amount as share-based compensation expense, even though share-based compensation is correctly added back in the reconciliation of net income to cash flow from operating activities on the statement of cash flow. Your colleague has *most likely* forgotten to:

A. add share-based compensation expense to equity.

B. subtract share-based compensation expense from equity.

C. subtract the fair value of share-based award settlements from equity.

Feedback

A is correct. The offsetting entry to share-based compensation expense is to share-based compensation reserve, an account in equity. Ignoring this results in an expense which decreases retained earnings without a decrease in an asset or increase in a liability, thus causing an imbalance.

---

28 Multiple Choice 0 / 1 point

The subject company reported a net loss on its income statement in the most recent fiscal period but reported positive free cash flow. Your colleague should add anti-dilutive securities to diluted shares outstanding in their discounted cash flow valuation model to account for:

A. forfeited share-based awards.

Correct Answer: **B. unvested share-based awards.**

B. unvested share-based awards.

C. RSUs that settled in the most recent fiscal period.

Feedback

B is correct. All unvested share-based awards, regardless of the spread between their grant-date fair value and current market price, are anti-dilutive for a company that reported a net loss.

---

29 Multiple Choice 0 / 1 point

As an alternative to expensing stock-based compensation in free cash flow in the discounted cash flow valuation model, your colleague could account for the expected dilution by:

A. increasing shares outstanding.

B. reducing operating expenses because the company could compensate its employees in cash rather than shares.

C. increasing net cash because share-based compensation expense is a non-cash expense.

Correct Answer: **A. increasing shares outstanding.**

Feedback

A is correct. Dilution from share-based compensation can be accounted for by expensing share-based compensation in free cash flow or by increasing the share count for expected dilution.

B and C are incorrect as they would be accretive, not dilutive, to valuation.

---

30 Multiple Choice 1 / 1 point

The subject company's management has provided guidance for the next fiscal year, which includes an effective tax rate that is substantially below its statutory tax rate, largely due to tax windfalls from share-based compensation. Using this effective tax rate for all future years in the model would likely:

A. overstate income tax expense.

B. overstate free cash flow to the firm.

C. understate estimated enterprise value.

Feedback

B is correct. A lower effective tax would increase free cash flow to the firm (After-tax EBIT + D&A – Capex +/- Change in working capital). Tax windfalls or excess tax benefits decrease the effective tax rate versus the statutory rate because they are deductible for tax purposes, but not expensed on the income statement.